

Quick Check

Lesson 4: Tangent and the Sine–Cosine–Tangent Relationship: Solving the Mango-Tree Problem

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Trigonometry I | Lesson 4

A student in Kisumu measures the sun's angle with the ground as 53° . The shadow of a vertical mango tree on level ground is 9.0 m long. Which equation connects the angle, the shadow and the tree's height h directly?

A $\tan 53^\circ = 9.0 / h$

B $\tan 53^\circ = h / 9.0$

C $\cos 53^\circ = h / 9.0$

D $\sin 53^\circ = h / 9.0$

In a right-angled triangle, $\sin \theta = O/H$ and $\cos \theta = A/H$. What is $\sin \theta \div \cos \theta$, that is $(O/H) \div (A/H)$, after simplifying?

A $(O \times A) / H^2$

B O / H

C O / A

D A / O

From mathematical tables, $\sin 40^\circ = 0.6428$ and $\cos 40^\circ = 0.7660$. What is $\sin 40^\circ \div \cos 40^\circ$, to 3 decimal places, and which table value does it match?

A 1.41, which matches $\tan 40^\circ$

B 0.839, which matches $\tan 40^\circ$

C 1.19, which matches $\tan 50^\circ$

D 0.49, which matches $\tan 40^\circ$

We have proved that $\tan \theta = \sin \theta / \cos \theta$, so tan can be found from sin and cos. Why is the tangent ratio still useful?

- A** It is only useful when we know the hypotenuse
- B** It gives a direct route when we know an angle and one of the opposite or adjacent sides but not the hypotenuse
- C** It is only useful for angles of 53°
- D** It is not useful, because sin and cos can always replace it

The sun's rays make a 53° angle with level ground. A vertical tree of height h casts a 9.0 m shadow on the ground, forming a right-angled triangle. The 53° angle is at the tip of the shadow. Which side of the triangle is the 9.0 m shadow, relative to the 53° angle?

- A** The adjacent side, because it lies next to the 53° angle and is not the hypotenuse
- B** It cannot be named until the tree height is known
- C** The hypotenuse, because it is the length we measured
- D** The opposite side, because it lies on the ground

A mango tree casts a 9.0 m shadow when the sun's angle with the ground is 53° . Using $\tan 53^\circ \approx 1.3270$ from tables, what is the height of the tree, to 2 decimal places?

A 0.15 m

B 10.33 m

C 6.78 m

D 11.94 m

The same 53° sun angle at the same time of day gives the same tangent ratio for a small pole and for a tall tree. Why does $\tan \theta = \text{Opposite/Adjacent}$ work for triangles of any size?

- A Because the ratio Opposite/Adjacent stays constant for a given angle, whatever the size of the triangle
- B Because the ratio gets bigger as the shadow gets longer
- C Because the opposite side is always equal to the adjacent side
- D Because $\tan \theta$ depends on the length of the hypotenuse

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).